MILITARY SPECIFICATION

INDICATOR, COMPUTER CP-696( )/UD

This specification is approved for use by Electronics Research and Development Command, Department of the Army, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Computer Indicator CP-696( )/UD (reader), which shall have the necessary electrical and mechanical components to read the gamma and neutron doses of the glass and pin diodes, respectively, of Radiac Detector DT-236( )/PD. The gamma dosimeter shall be read by the flourometric method, using ultra-violet excitation. The neutron dosimeter shall be read by measuring of forward voltage drop of constant current under conditions of conductivity modulation.

2. APPLICABILITY

2.1 Issues of documents. The effective issue or revision of the following documents shall be that listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, unless (i) specific issues are set forth therefor in the cited specifications or (ii) issues different from those specified in the cited specifications are set forth in the solicitation. The date of the applicable DODISS and supplements thereto shall be as specified in the solicitation or contract.

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin
NN-P-71	- Pallet, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	- Strapping, Steel, and Seals

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-B-585	- Boxes, Wood, Wirebound
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood and Lock Corner
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple Wall
PPP-S-760	- Strapping, Nonmetallic (and Connectors)
PPP-T-97	- Tape; Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	- Marking of Electronic Items
MIL-F-14072	- Finishes for Ground Electronic Equipment
MIL-P-15024/6	- Plates, Identification, Equipment

STANDARDS

FEDERAL

FED-STD-101	- Preservation, Packaging, and Packing Materials, Test Procedures

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized and Containerized Unit Leads; 40 Inch x 48 Inch Pallets, Skids, Runner or Pallet Type Base
MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-415	- Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	- Maintainability Verification, Demonstration, Evaluation
MIL-STD-726	- Packaging Requirements Code
MIL-STD-781	- Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-810	- Environmental Test Methods

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2.2 OTHER PUBLICATIONS

ELECTRONICS COMMAND

EL-CP-5111-0002A - Nuclear Survivability Requirements for Detector, Radiac DT-236( )/PD and Computer Indicator CP-696( )/UD

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Design and construction; general. Except as otherwise specified herein, the detailed mechanical and electrical design of the equipment shall be accomplished by the contractor, subject to the requirements specified herein. Details of design not specified herein shall conform to good engineering practice. The government will furnish design data from previous efforts which the contractor may utilize for mechanical and electrical design information; however, the contractor shall be responsible for providing a completely engineered, fabricated, assembled and tested equipment in accordance with the requirements specified. The reader (CP-696( )/UD) shall be fully compatible with all DT-236( )/PD dosimeters. The reader shall provide necessary voltage to read forward voltage drop of the PIN neutron detector at constant current and to power the light system and light detecting system for the phosphate glass gamma sensitive element. Suitable means shall be provided for inserting and removing the dosimeter from the reader.

3.1.1 Gamma reading circuit. The gamma reading circuit shall consist of an ultraviolet source, optical filters, and a photodiode or miniature photomultiplier tube with the necessary associated electronics. A nuclear hard internal standard shall be used to check reader operation and calibration. It shall be capable of checking the entire gamma channel. If a pulse Xenon flash tube is used as the gamma channel U.V. light source, it shall be an end window probe stabilized tube such as an EG&G FX-201 or equivalent.

3.1.2 Neutron reading circuit. The neutron reading circuit shall consist of necessary electronics to read the forward voltage drop of the PIN diodes at constant current. A pulsed circuit shall be used to eliminate diode heating effects. A nuclear hard internal standard shall be used to check the reader operation and calibration. It shall be capable of checking the entire neutron channel.

3.1.3 Readout. Neutron and gamma doses shall be displayed using digital readout. Provision to read neutron and gamma doses separately or combined may be used.

3.1.4 Reader warm-up. Specified performance characteristics shall be obtainable 5 minutes after the reader is turned on.

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3.1.5 Operating range. The CP-696( )/UD shall measure and indicate the dose of the DT-236( )/PD over the range from 0-1000 rads. The minimum scale increment allowable shall be 1 rad and the maximum allowable shall be 25 rads.

3.1.6 Calibration check. Each CP-696( )/UD shall have the capability of checking the calibration before measurement of DT-236( )/PD is started. This can be internal to the reader or may be done with external calibration dosimeters. Reference dosimeters shall be used to allow calibration of the reader in the field. Each calibration set shall consist of a number of dosimeters sufficient to check reader calibration in both the neutron and gamma channels over the range of 0-1000 rads. The reference dosimeters shall contain Toshiba FD-7 glass or equivalent dosed to 20,000 rads with neutral density filters or mechanical masks to attenuate the light output to give equivalent gamma readings from 0-1000 rads in the calibration set. Precision resistors shall be used in place of the neutron diode to give equivalent neutron readings from 0-1000 rads in the calibration set.

3.1.7 Reader power supply. Power for the reader shall be supplied, where possible, by local sources of 220 VAC or 115 VAC, 50 or 60 Hz or, when unavailable, by self-contained batteries, either internal to the reader or in a separate battery package. The batteries shall be as assigned by the contracting officer. The batteries shall be able to be removed without dismantling the instrument. The reader shall also have the capability to be operated from 24 VDC vehicular power. The reader design shall provide protection from vehicle/charging system transients.

3.1.8 Reader controls. The reader shall be provided with circuit check, battery check, AC "ON, OFF" and other controls as required to operate the instrument and as approved by the design activity. The number of controls shall be kept to a minimum, however, and located so as to provide simple, facile, and maximum efficiency of operation.

3.1.9 Carrying case or haversack. The reader shall be housed in a case or a haversack capable of meeting specified service conditions. The case shall accommodate all accessories normally required for field use. A spring-loaded fold-down handle or carrying strap designed for comfortable handling and good balance shall be provided. Protection shall be provided for all external hardware. A pressure relief valve shall be provided.

3.2 First article. When specified in the contract, the contractor shall furnish first article units in accordance with 4.3 (see 6.2g).

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein, shall apply.

3.3.1 Microelectronics. Maximum use shall be made of large scale integration (LSI) of microelectronics consistent with the overall equipment requirements specified herein in order to fully exploit the performance, reliability, cost, size, weight, logistics and maintenance advantages of these techniques.

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3.3.2 One-part printed circuit connectors. One-part printed circuit connectors shall not be used.

3.3.3 Bonds and grounds. Bonding for equipment current path returns, RF potentials and shock hazards shall be so installed that expansion, contraction, or movement incident to normal service use and maintenance will not break or loosen the connection. Surface preparation for bonds and grounds shall be accomplished by removing all anodic film, grease, paint, lacquer or other high resistance coatings from the immediate areas of contact. The equipment finish shall be restored to its original condition in accordance with MIL-F-14072. The DC resistance of bonds and grounds shall not exceed 2.5 milliohms.

3.3.4 Cables and connectors. Shielded cables, when required, shall have a shield weave providing not less than ninety percent coverage. The shield shall be constructed of braided, tinned copper wire. All shielded cables shall have the shield terminated in an approved military connector with an EMI back shell to provide peripheral bonding of the shield. The use of shielded cables shall be kept to a minimum by the use of alternative suppression techniques such as pin filters. All panel mounted connectors and components and the panel mating surfaces shall have a RF conductive finish and provide positive bonding and grounding with mating connectors and equipment ground plane. The DC bonding resistance for all connectors and panel mounted components shall not exceed 2.5 milliohms.

3.4 Finish. All surfaces of the CP-696( )/UD requiring a protective coating shall be finished in accordance with MIL-F-14072 (see 4.4).

3.5 Nameplate and product marking. (see 4.4)

3.5.1 General. Marking shall be in accordance with MIL-M-13231 (see 4.4). Front panel markings shall be group I as described in that specification.

3.5.2 Nameplates. Each nomenclatured item shall have a nameplate conforming to MIL-P-15024/6.

3.5.3 Serial numbers. Each CP-696( )/UD shall be serial numbered.

3.6 Interchangeability. Like units, subassemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454 (see 4.6).

3.7 Burn-in. Each CP-696( )/UD shall meet the requirements herein, without subsequent processing, after subjection to the burn-in of 4.7.

3.8 Size and weight. The weight of the reader including the carrying case shall not exceed 25 pounds (11 kg). No external dimension shall exceed 14 inches (35 cm) (see 4.8).

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3.9 Safety engineering.

3.9.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454. Compliance with these requirements will be verified through a visual inspection (see 4.14).

3.9.2 Corner and edge rounding. All exposed corners and edges shall be rounded to eliminate possible injury to personnel due to lacerations and cuts.

3.9.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses).

3.10 Performance requirements.

3.10.1 Accuracy. The CP-696( )/UD, when reading DT-236( )/PD dosimeters, shall have an accuracy of ± 20 percent or ± 10 rads of the true dose delivered to the dosimeters over the temperature range of -25°F (-32°C) to +125°F (+52°C) (see 4.9.1).

3.10.2 Precision. Statistically valid samples of DT-236( )/PD's shall give readings to within ± 5 percent of their own mean with a 95 percent confidence level when dosed to 500 rads and read repeatedly on the CP-696( )/UD (see 4.9.2).

3.10.3 Reading rate. The CP-696( )/UD shall be designed to allow reading at least 60 DT-236( )/PD's per hour (see 4.9.3).

3.10.4 Voltage range. The CP-696( )/UD shall be operable between 21 to 30 volts DC (see 4.9.4).

3.10.5 Orientation. The reader shall be operable in any orientation up to 20° from the normal operating position (see 4.9.5).

3.11 Service conditions. The CP-696( )/UD shall be subjected to and meet the service conditions below. For the purpose of service conditions, specified performance is defined as meeting the following requirements:

Requirement paragraph	Test paragraph	Measurement
3.10.1	4.9.1	Accuracy
3.10.2	4.9.2	Precision
3.10.3	4.9.3	Reading rate
3.10.5	4.9.5	Orientation

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3.11.1 High temperature.

a. Operating. The CP-696( )/UD shall be operable without degradation in specified performance at ambient test temperature as high as +125°F(+52°C).

b. Storage and transportation. The CP-696( )/UD shall withstand exposure to ambient air temperature as high as +160°F(+71°C) (see 4.10.1).

3.11.2 Low temperature.

a. Operating. The CP-696( )/UD shall be operable without degradation in specified performance at ambient temperature as low as -25°F(-32°C).

b. Storage and transportation. The CP-696( )/UD shall withstand exposure to ambient air temperature as low as -70°F(-57°C) (see 4.10.2).

3.11.3 Humidity. The CP-696( )/UD shall exhibit no physical damage such as corrosion, rust, blistering, swelling, or deterioration of parts and materials and shall meet the specified performance when subjected to 94 ± 4 percent relative humidity. The reader shall meet specified performance without degradation after completion of the test (see 4.10.3).

3.11.4 Altitude. The CP-696( )/UD shall meet the specified performance at 10,000 ft (3048 m) above sea level. The reader shall meet specified performance without degradation after completion of the test (see 4.10.4).

3.11.5 Immersion. The CP-696( )/UD as prepared for field transportation shall show no evidence of leakage after being tested in accordance with 4.10.5.

3.11.6 Shock, transit drop. The CP-696( )/UD shall withstand shocks encountered during operation (see 4.10.6).

3.11.7 Vibration. The CP-696( )/UD shall operate without degradation in specified performance and shall suffer no mechanical damage after being subjected to the vibration test specified in 4.10.7.

3.11.8 Shock, bench handling. The CP-696( )/UD shall operate without degradation in specified performance and shall suffer no mechanical damage after being subjected to the bench handling test specified in 4.10.8.

3.11.9 Dust. The CP-696( )/UD shall operate without degradation in specified performance after being subjected to the dust test of 4.10.9.

3.11.10 Salt fog. After the salt fog test of 4.10.10, the CP-696( )/UD shall operate without degradation in specified performance and shall show no evidence of degradation such as flaking, pitting, blistering or loosening of finish on metal surface or exfoliation (see 6.5). There shall also be no binding or clogging of mechanical parts.

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3.11.11 Fungus. After the fungus test of 4.10.11, the CP-696( )/UD shall be visually examined. The reader shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10 percent or less of the total area and no more than 6 unrelated minute colonies with mycelial development in areas only in other than critical circuit portions; such as terminal spacing, printed circuit boards, etc., with sparse growth due to random contamination or traces of unmixed material ingredients. In addition, the reader shall operate without degradation in specified performance after the test of 4.10.11. Materials used in the manufacture, fabrication, construction and treatment of the CP-696( )/UD shall comply with Requirement 4 of MIL-STD-454.

3.11.12 Temperature shock. The equipment shall be capable of withstanding temperature shocks between -65°F (-54°C) and +145°F (63°C) (see 4.10.12).

3.11.13 Nuclear survivability. The CP-696( )/UD shall meet the specified performance after being subjected to the nuclear environment of 4.10.13.

3.12 Electromagnetic compatibility. The reader shall meet the following emission and susceptibility requirements of MIL-STD-461, Notice 4 (see 4.11).

CE01	CS01	RE02	RS03 1/
CE02	CS02	RE02.1	
CE04	CS06		

1/ The frequency range of test shall be 10 kHz to 10 GHz with the equipment immersed in the following radiated fields:

10	kHz	-	1.9 MHz	1 volt/meter field
2.0	MHz	-	29.99 MHz	5 volt/meter field
30.0	MHz	-	1999 MHz	10 volt/meter field
2000	MHz	-	10 GHz	5 volt/meter field

All electromagnetic interference (EMI) tests shall be performed for both AC power and DC power operations.

3.13 Reliability. The CP-696( )/UD shall have a specified (Θ_0) mean-time-between-failure (MTBF) of 800 hours. The lower test MTBF (Θ_1) shall be 400 hours (see 4.12). The reliability requirements shall be demonstrated in accordance with MIL-STD-781 when operating under the following conditions:

a. Operating temperature: -25°F to +125°F
(-32°C to +52°C)

b. Environment: ground mobile

c. Duty cycle: 100 percent

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3.14 Maintainability. Optimum accessibility for all components requiring maintenance inspection, removal or replacement shall be provided. Test points shall be incorporated to permit fault diagnostic procedures and repair, alignment and calibration. Test points shall conform to MIL-STD-415. Mean-time-to-repair shall not exceed 5 minutes at organizational support and 40 minutes at general support. Maximum-time-to-repair shall not exceed 10 minutes at organizational support and 2 hours at general support (see 4.13).

3.15 Recycled, virgin and reclaimed materials. It is required that, to the maximum extent possible, recovered or reclaimed materials, in lieu of the virgin materials, shall be used without jeopardizing the intended use of the manufactured item.

3.16 Workmanship. The CP-696( )/UD shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of MIL-P-11268.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3). Does not include packaging.

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspections:

Quality conformance inspection of equipment before packaging (see 4.5).

Quality conformance inspection of packaging (see 4.16).

4.3 First article. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor (see 3.2).

4.3.1 First article units. The contractor shall furnish 12 first article units of the CP-696( )/UD.

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4.3.2 First article inspection. The first article inspection shall consist of the inspection specified in table I and shall be performed in the order specified.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirements specified in the contract.

Item	Where required
Parts, materials, and processes; general	3.3
Finish	3.4
Marking	3.5
Workmanship	3.16

TABLE I. First article inspection.

Inspections	Reqt para	Test para	Order of tests 2/				
			Unit 1	Unit 2	Unit 3	Units 4-7	Units 8-12
1. Inspections covered by subsidiary documents 1/	3.3 3.4 3.5 3.16	4.4	Inspection to be performed on all units				
2. Burn-in 1/	3.7	4.7	Inspection to be performed on all units				
3. Group A inspection 1/	See table II		Inspection to be performed on all units				
4. Group B inspection 1/	See table III		Inspection to be performed on all units				

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TABLE I. First article inspection (continued).

Inspections	Reqt para	Test para	Order of tests 2/				
			Unit 1	Unit 2	Unit 3	Units 4-7	Units 8-12
5. Group C inspection							
High temperature	3.11.1	4.10.1	1				
Low temperature	3.11.2	4.10.2	2				
Humidity	3.11.3	4.10.3	4				
Immersion	3.11.5	4.10.5		3			
Dust	3.11.9	4.10.9		2			
Vibration	3.11.7	4.10.7			1		
Shock, transit drop	3.11.6	4.10.6			2		
Shock, bench handling	3.11.8	4.10.8			3		
Salt fog	3.11.10	4.10.10		4			
Fungus	3.11.11	4.10.11			4		
Altitude	3.11.4	4.10.4		1			
Temperature shock	3.11.12	4.10.12	3				
6. Unique first article tests							
Reading rate 3/	3.10.3	4.9.3				1	
Maintainability	3.14	4.13				2	
Nuclear survivability	3.11.13	4.10.13					
Nuclear air blast	3.11.13	4.10.13.3				Unit 4	
Thermal radiation	3.11.13	4.10.13.2				Unit 5	
Electromagnetic pulse	3.11.13	4.10.13.1				Unit 6	
Initial nuclear radiation	3.11.13	4.10.13.4				Unit 6	
Electromagnetic compatibility	3.12	4.11				Unit 7	
7. Group D Reliability	3.13	4.12					1

1/ The inspection 1 to 4, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements.

2/ The numbers in the unit columns in the table, covered by inspections 5 and 6, specify the order of inspections for the indicated units. Each of the 4 nuclear survivability tests shall be performed on one unit only after tests 1 and 2. The electromagnetic pulse test will be performed on unit 6 before the initial nuclear radiation test.

3/ This test involves human performance reliability and equipment compatibility.

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4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract shall be inspected for conformance to the inspections specified in table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, nor in the contract shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels and AQLs indicated in table II.

4.5.1.1 Order of inspection within group A. A group A inspection shall be performed in an order satisfactory to the government.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test paragraph	AQL major	AQL minor
Accuracy	3.10.1	4.9.1	1%	1/
Precision	3.10.2	4.9.2	1%	
Orientation	3.10.5	4.9.5	1%	
Visual and mechanical	3.1	4.15	1%	4%
Safety engineering	3.9	4.14	1%	4%

1/ All performance defects are major.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to table III and to the special inspection levels of table I of MIL-STD-105.

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4.5.2.1 Group B sampling plans. The group B sample plans, for the AQLs listed in table III, shall be inspection level S-4.

4.5.2.2 Order of inspection within group B. Group B inspection shall be performed in an order satisfactory to the government.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test paragraph	AQL
Interchangeability	3.6	4.6	6.5%
Size and weight	3.8	4.8	4%
Voltage range	3.10.4	4.9.4	1%

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for group C inspection. Four samples of CP-696( )/UD shall be selected at random, one sample for each subgroup in table IV from the first 400 units or fraction thereof produced. The electromagnetic compatibility test shall be performed only on the first production lot. Thereafter, 3 samples of CP-696( )/UD shall be selected at random from every 400 units or fraction thereof greater or equal to 200 units, one unit for each subgroup 1, 2 and 3 of table IV. If the fraction is less than 200, the selection of group C samples for the preceding test shall include units from the remaining fraction.

4.5.3.2 Group C failures. Action required relative to group C failure shall be as specified in the contract.

4.5.3.3 Reinspection of conforming group C sample units. *Unless otherwise* specified, sample units which have been subjected to and passed group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group B and group A inspection, in that order.

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TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
High temperature	3.11.1	4.10.1
Low temperature	3.11.2	4.10.2
Temperature shock	3.11.12	4.10.12
Humidity	3.11.3	4.10.3
Subgroup 2		
Altitude	3.11.4	4.10.4
Dust	3.11.9	4.10.9
Salt fog 1/	3.11.10	4.10.10
Subgroup 3		
Vibration	3.11.7	4.10.7
Shock, transit drop	3.11.6	4.10.6
Bench handling	3.11.8	4.10.8
Immersion	3.11.5	4.10.5
Fungus 1/	3.11.11	4.10.11
Subgroup 4		
Electromagnetic compatibility	3.12	4.11

1/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

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4.5.4 Group D inspection. This inspection shall consist of the tests specified in table V and shall be performed on units from lots which have been subjected to and met group A and group B inspection.

4.5.4.1 Sampling for group D inspection. Five (5) units shall be selected at random from the first 400 units or fraction thereof and subjected to the test of 4.12.

4.5.4.2 Group D failures. Actions relative to group D failures shall be as specified in the contract.

TABLE V. Group D inspection.

Inspection	Requirement paragraph	Test paragraph
Reliability	3.13	4.12

4.5.4.3 Reinspection of conforming group D sample units. Unless otherwise specified, sample units which have been subjected to and passed group D inspection may be accepted on contract provided all visible and known damage is repaired and the sample units are resubjected to and pass group A inspection.

4.6 Interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.6. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units, and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.7 Burn-in. Each CP-696( )/UD shall be operated for a minimum of 24 hours under the test conditions specified in 4.12. The burn-in shall be concluded by a failure free period, the duration of which shall be 12 hours. Equipment performance shall be monitored as specified in 4.12 (see 3.7).

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4.8 Size and weight.

4.8.1 Size. The size of the CP-696( )/UD shall be measured. It shall meet the requirements of 3.8.

4.8.2 Weight. The CP-696( )/UD shall be weighted. The weight shall not exceed that given in 3.8.

4.9 Performance tests.

4.9.1 Accuracy. The DT-236( )/PD dosimeters exposed to 50, 100, 200, 300, 400, 500, 600, 700, 800, 900 and 1000 rads gamma and neutron respectively shall be read at -25°F, 0°F, 32°F, 68°F and 125°F and shall meet the accuracy requirements given in 3.10.1.

4.9.2 Precision. A minimum of 10 DT-236( )/PD's shall be exposed to 500 rads neutron and 500 rads gamma. The mean for gamma and neutron dose shall be calculated. Five subsequent readings for gamma and neutron shall be taken (see 3.10.2).

4.9.3 Reading rate. Fifteen dosimeters shall be opened, read and resealed in less than 15 minutes (see 3.10.3).

4.9.4 Voltage range. The DT-236( )/PD dosimeters exposed to 50, 100, 200, 300, 400, 500, 700, 800, 900 and 1000 rads gamma and neutron respectively shall be read at -25°F and 125°F. The input voltage to the reader shall be set at 21 volts and 30 volts and readings taken at both temperatures. Accuracy shall be in accordance with 3.10.1 (see 3.10.4).

4.9.5 Orientation. The dosimeters shall be read with the reader set at 20° from the normal operating position tilted back, front, and to both sides (see 3.10.5).

4.10 Service conditions. The service condition tests shall be performed as detailed below. Each CP-696( )/UD subjected to these tests shall have passed group A and B tests and shall be resubmitted to and pass the group A tests after completion of all service condition tests. Where operation of the test item is required during any service condition test, the test item shall be required to pass the specified performance.

4.10.1 High temperature. The CP-696( )/UD shall be subjected to the test of Method 501.1, Procedure II of MIL-STD-810. The storage temperature (Step 4) shall be 160°F (71°C). In Step 7, the operating temperature shall be +125°F (+52°C) (see 3.11.1).

4.10.2 Low temperature. The CP-696( )/UD shall be subjected to the test of Method 502.1, Procedure I of MIL-STD-810. The storage temperature (Step 2) shall be -70°F (-57°C) and shall be maintained for a period of not less than 2 hours following stabilization of the test item. Low operating temperature (Step 4) shall be -25°F (-32°C) (see 3.11.2).

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4.10.3 Humidity. The CP-696( )/UD shall be subjected to the test of Method 507.1, Procedure III of MIL-STD-810. In Step 6, measurements shall be taken during the last 5 hours of the last cycle (see 3.11.3).

4.10.4 Altitude. The CP-696( )/UD shall be subjected to the test of Method 500.1, Procedure I of MIL-STD-810. The equipment shall be held at 40,000 feet (1219 m) non-operating for one hour. The operating altitude in Step 2 shall be 10,000 feet (3,048 m) (see 3.11.4).

4.10.5 Immersion. The CP-696( )/UD shall be subjected to the test of Method 512.1, Procedure I of MIL-STD-810 (see 3.11.5).

4.10.6 Shock, transit drop. The CP-696( )/UD shall be subjected to the test of Method 516.2, Procedure II of MIL-STD-810.

4.10.7 Vibration. The CP-696( )/U in its carrying case shall be subjected to the test of Method 514.2, Procedure X of MIL-STD-810 (see 3.11.7).

4.10.8 Shock, bench handling. The CP-696( )/UD shall be subjected to the test of Method 516.2, Procedure V of MIL-STD-810 (see 3.11.8).

4.10.9 Dust. The CP-696( )/UD shall be subjected to the test of Method 510.1, Procedure I of MIL-STD-810 (see 3.11.9).

4.10.10 Salt fog. The CP-696( )/UD shall be subjected to the test of Method 409.1, Procedure I of MIL-STD-810 (see 3.11.10).

4.10.11 Fungus. The CP-696( )/UD as prepared for operation, shall be subjected to the test of Method 508.1, Procedure I of MIL-STD-810. There shall be abundant growth colonization on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the reader is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the reader (see 3.11.11).

4.10.12 Temperature shock. Each equipment shall be subjected to Test Method 503.1, Procedure I of MIL-STD-810 except that the high temperature shall be +145°F (63°C) (see 3.11.12).

4.10.13 Nuclear survivability. The CP-696( )/UD shall be subjected to the nuclear environment as specified in EL-CP5111-0002A. The reader shall be energized during exposure (see 3.11.13).

4.10.13.1 Electromagnetic pulse (EMP). The reader shall be placed within a long wire electromagnetic pulse facility or its equivalent in the following orientations:

a. Power cables extended parallel to the electric field and perpendicular to the magnetic field; all individual units under the long wire, but widely separated.

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b. Power cables coiled on ground; all units centrally located under the long wire.

Test data to satisfy the EMP requirements in EL-CP5111-0002A may be assembled by extrapolation (see 3.11.13).

4.10.13.2 Thermal radiation. One reader shall be exposed to the thermal radiation levels specified in EL-CP5111-0002A. Since the available simulation facilities provide these thermal radiation levels only over relatively small areas, many individual exposures will be required so that each of the exposed surfaces (case, meters, knobs, cables, and other outboarded components) will receive the required thermal exposure. The reader, following exposures, shall show no degradation and shall meet specified performance (see 3.11.13).

4.10.13.3 Nuclear air blast. The reader shall be placed in a nuclear air blast simulation facility and exposed to the air blast environment as specified in EL-CP5111-0002A. The reader shall meet the specified performance following this test (see 3.11.13).

4.10.13.4 Initial nuclear radiation. The complete reader shall be exposed to a LINAC and a pulsed or fast burst reactor in such a manner that all electronic piece parts and circuits in the set receive the neutron and gamma doses and rates specified in EL-CP5111-0002A. The reader shall meet specified performance (see 3.11.13).

4.11 Electromagnetic compatibility. The reader shall be tested for compliance with the requirements of 3.12, using the approved test methods of MIL-STD-462.

4.12 Reliability testing.

4.12.1 First article and quality conformance group D inspection. To determine compliance with 3.13 during first article testing, the equipment shall be tested in accordance with test plan IIIC of MIL-STD-781. To determine compliance with 3.13 during quality conformance inspection, the equipment shall be tested in accordance with test plan IVC of MIL-STD-781. The equipment shall be tested in accordance with MIL-STD-781 and the government approved test plan and procedure as required in the contract. The sample size shall be 5 units.

4.12.2 Reliability test conditions. The test environment for the reliability test shall be under the following conditions:

a. Electrical stress. The equipment shall be operated at nominal design input voltage for 50 percent of the time and at maximum and minimum input voltage for 25 percent of the time, respectively. The maximum and minimum voltages shall be $\pm$10 percent of nominal.

b. Vibration stress. The equipment shall be vibrated 15 minutes each operating hour at 2.2 $\pm$ 0.22g peak acceleration at any non-resonant frequency between 20 and 60 Hz measured at the mounting point of the unit. Total vibration time in a 24 hour period shall be at least 4 hours, 2 hours at low temperature and 2 hours at high temperature.

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c. Thermal stress. The CP-696( )/UD shall be continuously operated under the following temperature cycling:

Transit	Ambient to -250F(-320C)	1 hour
Stabilized	-250F(-320C)	9 hours
Transit	-250F(-320C) to +1250F(+520C)	2 hours
Stabilized	+1250F(+520C)	9 hours
Transit	+1250F(+520C) to ambient	1 hour
Stabilized	Ambient	2 hours

d. Test cycle. The test cycle shall be 24 hours long.

e. Failure definition. A failure is defined as the inability of the equipment to perform its required function. This includes any events which causes departure from performance as required by this specification and approved test procedures.

f. Parameters to be measured. Performance parameters are to be measured and recorded at least once per day on each equipment during the reliability test in accordance with paragraph 5.5.3 of MIL-STD-781. A complete set of measurements in accordance with paragraphs 3.10.1 and 3.10.2 shall be taken daily during both the high and low temperature stabilization.

4.13 Maintainability demonstration. Compliance with the maintainability requirement ( ) shall be verified by conducting a phase II demonstration in accordance with the requirements of MIL-STD-471, Method 9 with a consumer risk ( ) of 10 percent. Twenty (20) maintenance tasks shall be selected and simulated. The tasks shall be 25 percent organizational and 75 percent direct/general support (see 3.14).

4.14 Safety engineering inspection. A visual inspection shall be performed in the presence of a government safety representative to determine that all requirements of 3.9 have been incorporated in the equipment. Inability to meet these requirements shall constitute a failure of the test. Results of this inspection shall be included in the Safety Assessment Report, if required in the contract.

4.15 Visual and mechanical inspection. These inspections shall be performed in any order which is satisfactory to the government. The units shall be examined for the applicable defects listed in MIL-STD-252 (see 3.9).

4.16 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of section 5.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or B as specified (see 6.2b).

5.1.1 Level A.

5.1.1.1 Cleaning. Each reader and haversack shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each reader and haversack shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservative application. No preservative required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods and submethods prescribed in MIL-P-116 as specified herein, insuring compliance with the applicable requirements of that specification.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed IC-1.

5.1.1.4.2 Reader. Each reader shall be unit packed as specified in MIL-STD-726, coded as follows: 1Ø-1-1-ØØ-00-NS-X-ED-Ø-00-A.

5.1.1.4.3 Haversack. Each haversack (if carrying case is not used) shall be unit packed method III as follows: secure each flap by means of the fasteners provided and neatly fold each haversack into a compact unit. Place the folded unit within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Closure shall be in accordance with the appendix of the box specification.

5.1.1.4.4 Consolidation. Consolidate each reader and haversack, unit packed as specified above, within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Closure shall be in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.2 Packing. Packing shall be level A or B, as specified (see 6.2b). Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents when tested in accordance with Methods 5016 and 5017 of FED-STD-101.

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5.2.1 Level A. A multiple quantity of reader and haversack, consolidated as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A. When a single unit is shipped to a single destination, the unit shall be individually packed as specified above.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination exceed either a total of 250 pounds (excluding the pallet) or a volume of 20 cubic feet; and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of container, packed as specified in 5.2.1 except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV. Stretch wrap in accordance with MIL-STD-147 is authorized for shipments within the continental United States and for containerized shipments.

5.2.2 Level B. A multiple quantity of reader and haversack, consolidated as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-640, class 2, style E or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more an the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film conforming to L-P-378, type IV. When a single unit is shipped to a single destination, no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination exceed either a total of 250 pounds (excluding the pallet) or a volume of 20 cubic feet, and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of reader and haversack, unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film conforming to L-P-378, type IV. Stretch wrap in accordance with MIL-STD-147 is authorized for shipments within the continental United States and for containerized shipments.

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5.3 Marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Computer Indicator CP-696( )/UD is intended to read gamma and neutron doses from Radiac Detector DT-236( )/PD in the field.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A or level B preservation and packing (see section 5).

c. When rough handling test is required (see section 4).

d. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

e. Marking and shipping of samples.

f. Place of final inspection.

g. Number of first articles to be submitted for approval (see 3.2).

6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Nomenclature. The parentheses in the nomenclature will be deleted and replaced by a letter identifying the particular design, for example CP-696A/PD. The contractor should apply for nomenclature in accordance with applicable clause in the contract.

6.5 Definitions. Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-I-49422(ER) INDICATOR, COMPUTER CP-696( )/UD

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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